Three-Dimensional Choreographic Orbits: Mathematical Framework

MathFrame — v0.1 — Companion to RDD v0.1 and CompSpec v0.1

e5bed97b3349 • March 2026

Abstract

This document develops the mathematical foundations underlying the three-dimensional choreographic orbit classification programme. We establish the variational framework, derive the Fourier-mode constraints imposed by each relevant three-dimensional point group via representation theory, state the symmetric-criticality existence theorem, develop the knot-theoretic characterisation of the spacetime trace, and present the Floquet–Krein stability theory for periodic Hamiltonian orbits. These results inform the research design and computational specification; they will also constitute the theory section of the eventual paper.

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1 Setup and Notation

1.1 Units and normalisation

Throughout we work in units where the gravitational constant $G = 1$, all masses $m = 1$, and the choreographic period $T = 1$. Distances are measured in units such that the characteristic action $\mathcal{A}_0 = m^2 G/T^{1/3} = 1$. Physical results are recovered by dimensional analysis.

1.2 Configuration space

The configuration space of N bodies in $\mathbb{R}^3$ is

$$X_N \;=\; (\mathbb{R}^3)^N \setminus \Delta, \qquad \Delta = \Big\{\mathbf{q} \in (\mathbb{R}^3)^N : \mathbf{q}_i = \mathbf{q}_j \text{ for some } i \neq j\Big\},$$

with Δ the collision set. The loop space we minimise over is

$$\Lambda_N \;=\; H^1(S^1,\, (\mathbb{R}^3)^N \setminus \Delta),$$

the Sobolev space of absolutely continuous closed paths avoiding collisions. The H^1 norm ensures the kinetic-energy term is finite and that Λ_N carries a Hilbert manifold structure.

1.3 The Newtonian action

The Lagrangian action for one period is

$$\mathcal{A}[\mathbf{q}_1, \ldots, \mathbf{q}_N] \;=\; \int_0^1 \left[\frac{1}{2} \sum_{i=1}^{N} |\dot{\mathbf{q}}_i|^2 \;+\; \sum_{i<j} \frac{1}{|\mathbf{q}_i - \mathbf{q}_j|} \right] dt. \tag{1}$$

Critical points of $\mathcal{A}$ on Λ_N are $T = 1$ periodic solutions of Newton's equations $\ddot{\mathbf{q}}_i = \sum_{j \neq i} (\mathbf{q}_j - \mathbf{q}_i) / |\mathbf{q}_j - \mathbf{q}_i|^3$.

2 The Choreographic Subspace

2.1 Definition and reduction

Definition 2.1 (Choreographic orbit)**.** A path $(\mathbf{q}_1, \ldots, \mathbf{q}_N) \in \Lambda_N$ is *choreographic* if there exists a single closed curve $\gamma \in H^1(S^1, \mathbb{R}^3)$ such that

$$\mathbf{q}_i(t) = \gamma\Big(t - \tfrac{i-1}{N}\Big), \qquad i = 1, \ldots, N,$$

where time is taken modulo 1. The bodies follow the same curve in sequence, each offset by phase $1/N$.

Substituting Definition 2.1 into (1) and using T-periodicity collapses the functional to a single-curve action:

$$\mathcal{A}[\gamma] \;=\; \frac{1}{2} \int_0^1 |\dot{\gamma}|^2 \, dt \;+\; \frac{1}{N} \sum_{k=1}^{N-1} \int_0^1 \frac{dt}{|\gamma(t) - \gamma(t + k/N)|}. \tag{2}$$

The choreographic subspace $\mathcal{C}_N \subset \Lambda_N$ is the image of the embedding $\gamma \mapsto (\mathbf{q}_1, \ldots, \mathbf{q}_N)$; it is a closed submanifold of Λ_N. Critical points of $\mathcal{A}|_{\mathcal{C}_N}$ correspond to genuine N-body solutions.

2.2 Fourier representation

Since $\gamma \in H^1(S^1, \mathbb{R}^3)$, it admits a Fourier expansion

$$\gamma(t) = \sum_{k=1}^{\infty} \left[\mathbf{a}_k \cos(2\pi k t) + \mathbf{b}_k \sin(2\pi k t) \right], \qquad \mathbf{a}_k, \mathbf{b}_k \in \mathbb{R}^3, \tag{3}$$

where the zero mode $\mathbf{a}_0 = 0$ by the centre-of-mass constraint $\sum_i \mathbf{q}_i \equiv 0$. The H^1 condition reads $\sum_{k=1}^{\infty} k^2 (|\mathbf{a}_k|^2 + |\mathbf{b}_k|^2) < \infty$.

It is convenient to pass to complex amplitudes. Write $\gamma(t) = \mathrm{Re} \sum_{k=1}^{\infty} \mathbf{z}_k \, e^{2\pi i k t}$, so $\mathbf{z}_k = \mathbf{a}_k - i\mathbf{b}_k \in \mathbb{C}^3$. The phase-shift $\gamma(t) \mapsto \gamma(t+s)$ acts by $\mathbf{z}_k \mapsto e^{2\pi i k s} \mathbf{z}_k$; the spatial transformation $R \in \mathrm{O}(3)$ acts by $\mathbf{z}_k \mapsto R\mathbf{z}_k$ (applied componentwise in $\mathbb{C}^3$ via the complexification of R).

3 Symmetry Groups and Equivariant Critical Points

3.1 The symmetry group of a choreography

The symmetry group of a choreographic orbit is a subgroup of

$$G \subset \mathrm{O}(3) \times D_N,$$

where $D_N = \mathbb{Z}/N\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$ is the dihedral group of order $2N$. The $\mathbb{Z}/N\mathbb{Z}$ factor permutes the bodies cyclically; the $\mathbb{Z}/2\mathbb{Z}$ factor is time-reversal $t \mapsto -t$, which for an equal-mass choreography acts as a spatial isometry composed with reversal of the traversal direction. A generator of the symmetry group is a triple (R, σ, ϵ) with $R \in \mathrm{O}(3)$, $\sigma \in \mathbb{Z}/N\mathbb{Z}$, $\epsilon \in \{+1, -1\}$, acting on γ by

$$(R, \sigma, \epsilon) \cdot \gamma(t) \;=\; R\,\gamma\Big(\epsilon t - \tfrac{\sigma}{N}\Big)\,. \tag{4}$$

Most groups arising in practice have $\epsilon = +1$ (no time-reversal); we specialise to this case initially and note where $\epsilon = -1$ is required.

3.2 Induced representation on Fourier modes

Applying (4) to the Fourier series (3) with $\epsilon = +1$ gives, for mode k:

$$\mathbf{z}_k \;\longmapsto\; e^{-2\pi i k\sigma/N}\, R\,\mathbf{z}_k. \tag{5}$$

For γ to be a fixed point of $(R, \sigma, +1)$, each mode must satisfy

$$\mathbf{z}_k \;=\; e^{-2\pi i k\sigma/N}\, R\,\mathbf{z}_k, \tag{6}$$

i.e. $\mathbf{z}_k$ lies in the $e^{2\pi i k\sigma/N}$-eigenspace of R (in the complexification of $\mathbb{R}^3$). This is the key equation from which all symmetry constraints follow.

3.3 The Principle of Symmetric Criticality

Theorem 3.1 (Palais, 1979)**.** *Let a compact Lie group G act smoothly and isometrically on a Hilbert manifold $\mathcal{M}$, and let $f : \mathcal{M} \to \mathbb{R}$ be a G-invariant C^1 functional. Then every critical point of $f|_{\mathrm{Fix}(G)}$ is a critical point of f on all of $\mathcal{M}$.*

Corollary 3.2. *Every local minimiser of $\mathcal{A}|_{\mathrm{Fix}(G)\cap\mathcal{C}_N}$ is a genuine N-body choreographic solution.*

The theorem applies here because $\mathrm{O}(3)$ is compact and acts isometrically on $H^1(S^1, \mathbb{R}^3)$ with respect to the natural L^2 inner product, and $\mathcal{A}$ is manifestly $\mathrm{O}(3)$-invariant. The Reynolds projector $P_G = |G|^{-1} \sum_{g\in G} \rho(g)$ is the orthogonal projection onto $\mathrm{Fix}(G)$.

4 Point-Group Constraints: Fourier Mode Analysis

We now apply (6) systematically to each point-group family. For each group we state: the generator(s), the surviving modes, and the dimension count. We use the Schönflies notation throughout.

4.1 Cyclic groups C_n

The group C_n is generated by $(R_n, 1, +1)$, where $R_n \in \mathrm{SO}(3)$ is rotation by $2\pi/n$ about the z-axis and $\sigma = 1$. The complexified eigenvalues of R_n are $\{e^{2\pi i/n}, e^{-2\pi i/n}, 1\}$ for the three components (z_+, z_-, z_0) in the helical basis, where $z_\pm = x \pm iy$ and $z_0 = z$.

Equation (6) for this generator becomes:

$$\mathbf{z}_k \;=\; e^{-2\pi i k/n}\, R_n\, \mathbf{z}_k.$$

Decomposing by helicity:

- **z_+-component** (eigenvalue $e^{2\pi i/n}$): condition is $e^{2\pi i/n} e^{-2\pi i k/n} = 1$, i.e. $n \mid (1-k)$, i.e. $k \equiv 1 \pmod{n}$.
- **z_--component** (eigenvalue $e^{-2\pi i/n}$): condition is $e^{-2\pi i/n} e^{-2\pi i k/n} = 1$, i.e. $n \mid (1+k)$, i.e. $k \equiv n-1 \pmod{n}$. Since $z_- = \overline{z_+}$, this is automatically satisfied when the z_+ condition holds.
- **z_0-component** (z-axis, eigenvalue 1): condition is $e^{-2\pi i k/n} = 1$, i.e. $n \mid k$, i.e. $k \equiv 0 \pmod{n}$.

Proposition 4.1 (C_n mode selection)**.** *For an $N = n$ body choreography with C_n symmetry, the surviving Fourier modes are:*

$$\textit{xy-plane}: \quad k \in \{1, n+1, 2n+1, \ldots\}, \qquad \textit{z-axis}: \quad k \in \{n, 2n, 3n, \ldots\}.$$

For a truncation at K modes, the dimension of $\mathrm{Fix}(C_n)$ *is* $\dim_{\mathbb{R}} = 4\lfloor K/n \rfloor + 2\lfloor (K-1)/n + 1 \rfloor \approx 6K/n$.

Example 4.2 (C_3, $N = 3$)**.** Surviving modes: xy at $k = 1, 4, 7, 10, \ldots$; z at $k = 3, 6, 9, \ldots$. At truncation $K = 12$: 4 xy-modes ($k = 1, 4, 7, 10$) contributing 8 real d.o.f., plus 4 z-modes ($k = 3, 6, 9, 12$) contributing 4 real d.o.f. Total: 12 real parameters vs. the unconstrained 36. Reduction factor: 3.

4.2 Dihedral groups D_n, D_{nh}, D_{nd}

D_n adds a C_2 rotation about an axis perpendicular to z, e.g. R_{2x} (rotation by π about x), with body permutation $\sigma = 0$ (no phase shift). In helicity components: $R_{2x} : z_+ \mapsto -\overline{z_-}$, $z_0 \mapsto -z_0$. The additional constraint on surviving C_n modes is:

- **z_+-component**: R_{2x} maps the $k \equiv 1$ modes to $k \equiv n-1$ modes. Compatibility requires $\mathbf{z}_k^+ = -\overline{\mathbf{z}_k^-}$, fixing the imaginary part of z_+ in terms of the real part. Net effect: one real parameter per surviving xy mode instead of two.
- **z-component**: R_{2x} maps $z_0 \mapsto -z_0$, so $k \equiv 0$ z-modes are forced to zero.

Proposition 4.3 (D_n mode selection)**.** *For D_n symmetry with $N = n$, the surviving modes are: xy-plane modes at $k \equiv 1 \pmod{n}$ with* one *real d.o.f. each; z-axis modes vanish entirely. The orbit is therefore confined to a plane: D_n choreographies with $N = n$ are generically planar.*

This result is significant: it means that searching for non-planar choreographies requires symmetry groups where the z-modes are *not* killed. The C_n and C_{nv} families are the natural targets for genuinely 3D orbits.

D_{nh} and D_{nd}: Adding a horizontal reflection σ_h (for D_{nh}) maps $z_0 \to -z_0$, again eliminating z-modes and enforcing planarity. D_{nd} adds an S_{2n} improper rotation; the analysis is similar and also yields planar orbits for generic N.

4.3 Cyclic groups with vertical reflection: C_{nv}

C_{nv} adds a vertical reflection σ_v through the xz-plane, which acts as $(x, y, z) \mapsto (x, -y, z)$, i.e. $z_+ \mapsto \overline{z_+}$, with $\sigma = 0$ (no phase shift). The C_n mode selection survives; the σ_v constraint additionally requires $\mathbf{z}_k^{xy}$ to be real in the helicity basis, fixing $\mathrm{Im}(\mathbf{z}_k^{xy}) = 0$. For the z-modes, σ_v acts trivially, leaving them free.

Proposition 4.4 (C_{nv} mode selection)**.** *For C_{nv} with $N = n$: xy-plane modes at $k \equiv 1 \pmod n$ each contribute one real d.o.f. (not two); z-axis modes at $k \equiv 0 \pmod n$ each contribute two real d.o.f. (both $\mathbf{a}_k^z$ and $\mathbf{b}_k^z$ are free). The orbits are generically* non-planar*.*

The C_{nv} family therefore provides the simplest non-planar choreographic orbits with non-trivial symmetry. The C_{3v} case with $N = 3$ is our primary Phase 1 target.

4.4 Tetrahedral group T and T_d

The tetrahedral group $T \cong A_4$ has order 12, generated by a C_3 rotation about the $(1, 1, 1)$-axis and a C_2 rotation about the z-axis. For an N-body choreography with T symmetry, N must be divisible by the order of the C_3 element in the body-permutation representation; the smallest natural choice is $N = 12$ (one body per element of T).

Applying (6) to all 12 generators of T acting simultaneously constrains the surviving modes. The key result is:

Proposition 4.5 (T mode selection, $N = 12$)**.** *For $N = 12$ with full tetrahedral symmetry T, the first surviving mode index is $k = 1$ (which carries a 3-dimensional real space of coefficients transforming as the standard representation of T). Subsequent survivors occur at $k \equiv 1 \pmod{12}$ for the xy-components and $k \equiv 0 \pmod{12}$ for the z-component (the A_1 representation). At truncation $K = 48$ this gives approximately $48/12 \times 6 \approx 24$ real parameters.*

T_d adds improper S_4 axes. The additional constraint selects among representations of T and may further eliminate z-modes, potentially restricting to planar orbits; this requires case-by-case analysis.

4.5 Octahedral group O and O_h

The octahedral group $O \cong S_4$ has order 24. For $N = 24$ (one body per group element), the analysis parallels the tetrahedral case. The mode constraints are sparser: surviving modes occur at $k \equiv 1 \pmod{24}$ for the xy-components, and

the first non-trivial z-mode is at $k = 24$. At truncation $K = 96$: approximately $96/24 \times 6 = 24$ real parameters — the same dimensionality as the tetrahedral case, despite the larger group, because the higher symmetry compensates for the larger N.

$O_h = O \times \mathbb{Z}/2\mathbb{Z}$ (adding inversion) tends to force additional pairings between modes and can eliminate the z-modes entirely depending on N.

4.6 Icosahedral group I and I_h

$I \cong A_5$ has order 60, making it the largest finite subgroup of SO(3). For $N = 60$, the mode constraints are: xy-components survive at $k \equiv 1 \pmod{60}$ and z-components at $k \equiv 0 \pmod{60}$. At truncation $K = 240$: $240/60 \times 6 = 24$ real parameters — the same order as tetrahedral and octahedral cases. The key open question is whether $N = 12$ (a proper subgroup arrangement) admits icosahedral-class choreographies.

Remark 4.6. The dimension count suggests that the search is no harder for icosahedral than tetrahedral symmetry at fixed truncation ratio K/N — the larger group buys just as much reduction as the larger N costs. This is the dimensional miracle that makes the icosahedral case computationally tractable.

4.7 Summary table

Group	**Order**	**Natural N**	**xy modes**	**z modes**	**3D?**
C_n	n	n	$k \equiv 1 \pmod{n}$	$k \equiv 0 \pmod{n}$	Yes
C_{nv}	$2n$	n	$k \equiv 1 \pmod{n}$	$k \equiv 0 \pmod{n}$	Yes
D_n	$2n$	n	$k \equiv 1 \pmod{n}$	none	No
D_{nh}	$4n$	n	$k \equiv 1 \pmod{n}$	none	No
T	12	12	$k \equiv 1 \pmod{12}$	$k \equiv 0 \pmod{12}$	Yes
T_d	24	12	(reduced)	(case-dep.)	TBD
O	24	24	$k \equiv 1 \pmod{24}$	$k \equiv 0 \pmod{24}$	Yes
O_h	48	24	(reduced)	(case-dep.)	TBD
I	60	60	$k \equiv 1 \pmod{60}$	$k \equiv 0 \pmod{60}$	Yes
I_h	120	60	(reduced)	(case-dep.)	TBD

5 Existence Theory

5.1 Minimisation and the coercivity argument

The action $\mathcal{A}$ in (2) is bounded below (the potential is positive) but not coercive on all of $\mathcal{C}_N$ due to the possibility of collisions making $\mathcal{A} \to +\infty$. The key technique, introduced by Chenciner–Montgomery [1], is to restrict minimisation to a homotopy class $[\gamma]_0 \in \pi_0(\Lambda_N^G)$ that *excludes collisions by topology.*

Theorem 5.1 (Chenciner–Montgomery, 2000)**.** *If the homotopy class $[\gamma_0] \in \pi_1(\mathcal{C}_N/G)$ has the property that every path in this class must pass through the collision set Δ*

to degenerate, then $\mathcal{A}$ achieves its infimum on $[\gamma_0] \cap \mathrm{Fix}(G)$, and the minimiser is a smooth choreographic orbit.

The original proof uses the Marchal lemma [2]: any orbit with a collision can be locally deformed to reduce the action while remaining in the same homotopy class, ruling out collisions as minimisers.

5.2 Symmetry-forced collision exclusion

For point groups G with sufficiently high symmetry, collision exclusion follows directly from the symmetry constraints without invoking Marchal's lemma.

Proposition 5.2. *If the symmetry group G contains elements $(R_k, k, +1)$ for all $k = 1, \ldots, N-1$ such that R_k has no fixed point in $\mathbb{R}^3$, then the fixed-point subspace $\mathrm{Fix}(G)$ is collision-free.*

Proof. At a collision between bodies i and j, we have $\mathbf{q}_i(t_0) = \mathbf{q}_j(t_0)$ at some time t_0. By the choreographic constraint, $\mathbf{q}_j(t) = \mathbf{q}_1(t + (j-i)/N)$, so the collision becomes $\gamma(t_0) = \gamma(t_0 + (j-i)/N)$. Under the symmetry $(R_{j-i}, j-i, +1)$, this point must satisfy $\gamma(t_0) = R_{j-i}\gamma(t_0)$, i.e. $\gamma(t_0)$ is a fixed point of R_{j-i}. If R_{j-i} has no fixed point (e.g. if it is a rotation by an angle $\neq 0, \pi$ in $\mathrm{SO}(3)$), no collision is possible. □

This is directly applicable to the C_n and C_{nv} families for odd $n \geq 3$: the rotation $R_k = R_n^k$ has no fixed point for $k \not\equiv 0 \pmod{n}$, so the odd-n choreographies in these families are automatically collision-free.

5.3 Open existence questions

The following cases are open and constitute part of the research programme:

1. Does the C_{3v} class with $N = 3$ admit a genuinely non-planar minimiser? (Numerically expected: yes.)
2. Do any tetrahedral-class ($N = 12$) minimisers exist in the $3D$-enabled modes?
3. Does the icosahedral case ($N = 60$) admit a minimiser, or does the small search space force the minimiser to collapse to a lower-symmetry orbit?

6 Knot Theory of the Spacetime Trace

6.1 The configuration space and its topology

The unordered configuration space of N distinct points in $\mathbb{R}^3$ is

$$\mathrm{UConf}_N(\mathbb{R}^3) \;=\; \Big((\mathbb{R}^3)^N \setminus \Delta\Big)/S_N.$$

The fundamental group $\pi_1(\mathrm{UConf}_N(\mathbb{R}^3))$ is the braid group B_N. A choreographic orbit traces a loop in $\mathrm{UConf}_N(\mathbb{R}^3)$ (since the bodies return to their starting configuration after one period), and its homotopy class is a conjugacy class in B_N — the topological type of the orbit.

6.2 From braid to knot: the spacetime trace

A choreographic orbit defines a *pure braid* in B_N: body 1 traces a closed curve and returns to its starting position (not permuted to another position). The spacetime trace of body 1 is the closed curve

$$\Gamma : S^1 \to \mathbb{R}^3, \qquad \Gamma(\phi) = \gamma\Big(\tfrac{\phi}{2\pi}\Big)\,, \tag{7}$$

where we close the time-axis $[0, 1]$ into S^1. Topologically, Γ is a knot (or link) in $S^3 = \mathbb{R}^3 \cup \{\infty\}$.

For planar choreographies, Γ lies in a solid torus (thickening of the orbital plane times S^1) and is classified by a braid word in B_N. For three-dimensional choreographies, Γ can leave the solid torus and realise genuinely knotted isotopy types not accessible from planar orbits.

Remark 6.1. The symmetry group G constrains the knot type of Γ. If $(R, 0, +1) \in G$ is a spatial symmetry (no body permutation), then R acts on Γ as a knot symmetry: $R \cdot \Gamma$ is isotopic to Γ. This places Γ in the sub-collection of knots admitting the symmetry R — a significant restriction on the knot table.

6.3 Polynomial invariants

The following invariants are computed for each discovered orbit.

Alexander polynomial. The Alexander polynomial $\Delta_\Gamma(t) \in \mathbb{Z}[t^{\pm 1}]$ is computed from the Seifert matrix V of Γ (a Seifert surface is triangulated from a regular projection):

$$\Delta_\Gamma(t) = \det\Big(t^{1/2}V - t^{-1/2}V^\top\Big)\,.$$

The Alexander polynomial detects the unknot (it equals 1) and distinguishes many knot types, but is not a complete invariant.

HOMFLY-PT polynomial. The HOMFLY-PT polynomial $P_\Gamma(v, z) \in \mathbb{Z}[v^{\pm 1}, z^{\pm 1}]$ satisfies the skein relation

$$v^{-1}P_{L_+} - vP_{L_-} = zP_{L_0},$$

where L_+, L_-, L_0 are diagrams differing at a single crossing. It strictly dominates the Alexander polynomial (and the Jones polynomial as a special case) and provides the finest easily-computable invariant available.

Hyperbolic volume. If the knot complement $S^3 \setminus \Gamma$ admits a hyperbolic metric (the generic case by Thurston's geometrisation theorem), its volume $\mathrm{vol}(\Gamma)$ is a complete knot invariant in the hyperbolic class. Computed via SnapPy.

6.4 The chemistry correspondence

The class of *torus knots* $T(p, q)$ (curves winding p times around the longitudinal direction and q times around the meridional direction of a torus) is of particular interest. Torus knots are the topological type of the most commonly synthesised

molecular knots: the trefoil $T(2,3)$ and the cinquefoil $T(2,5)$ have been realised as single-molecule structures [6].

We conjecture that C_n-symmetric choreographies with $n = 2p+1$ preferentially realise torus knots of type $T(2,n)$: the z-axis modes generate the "tube" direction of the torus while the xy-plane modes generate the winding. If confirmed, this establishes a direct map from symmetry class to knot type and makes the chemistry connection precise.

7 Floquet–Krein Stability Theory

7.1 Linearisation

Let $\mathbf{x}^*(t) = (\mathbf{q}^*(t), \dot{\mathbf{q}}^*(t)) \in T^*(\mathbb{R}^3)^N$ be a choreographic periodic orbit of period 1. The linearised equations of motion for a displacement $\delta\mathbf{x}(t)$ are

$$\frac{d}{dt}\delta\mathbf{x} \;=\; J(t)\,\delta\mathbf{x}, \qquad J(t) = \begin{pmatrix} 0 & I \\ D^2U(\mathbf{q}^*(t)) & 0 \end{pmatrix}, \tag{8}$$

where $U = -\sum_{i<j} |\mathbf{q}_i - \mathbf{q}_j|^{-1}$ is the potential and D^2U is its Hessian matrix. Equation (8) is a linear Hamiltonian system with periodic coefficients.

7.2 The monodromy matrix

The fundamental matrix $\Phi(t)$ of (8) satisfies $\dot{\Phi} = J(t)\Phi$, $\Phi(0) = I$. The *monodromy matrix* $M = \Phi(1)$ encodes the linearised dynamics over one period. Since (8) is Hamiltonian, $M \in \mathrm{Sp}(6N, \mathbb{R})$: it is symplectic, $M^\top \Omega M = \Omega$, where Ω is the standard symplectic form.

The eigenvalues of M occur in groups consistent with the symplectic structure:

- Eigenvalues on the unit circle come in conjugate pairs $\{e^{i\theta}, e^{-i\theta}\}$ (elliptic modes).
- Real eigenvalues come in reciprocal pairs $\{\lambda, \lambda^{-1}\}$ (hyperbolic modes).
- Off-axis eigenvalues come in quadruples $\{\lambda, \bar{\lambda}, \lambda^{-1}, \bar{\lambda}^{-1}\}$ (complex-hyperbolic modes).

By symmetry, $\lambda = 1$ always has multiplicity at least 4 (corresponding to time-translation and the three spatial translations). These are the *trivial* eigenvalues.

7.3 Krein signatures

For elliptic eigenvalues $e^{\pm i\theta}$, linear stability is not sufficient: one must also classify the Krein signature to determine whether the orbit can be destabilised by arbitrarily small perturbations. The Krein signature of an elliptic eigenvalue $\lambda = e^{i\theta}$ is the sign of the quadratic form $\Omega(\cdot, M\cdot)$ restricted to the λ-eigenspace.

Theorem 7.1 (Krein–Moser)**.** *An elliptic eigenvalue $e^{i\theta}$ of M is* Krein-stable *(i.e. cannot leave the unit circle under small perturbations of M within* $\mathrm{Sp}(6N, \mathbb{R})$*) if*

and only if its Krein signature is definite (all $+1$ or all -1). Instability requires the collision of two eigenvalues with opposite Krein signatures.

Corollary 7.2. *An orbit is* Krein-stable *if all its non-trivial elliptic eigenvalues have definite Krein signature. Krein stability implies linear stability and is a necessary condition for KAM non-linear stability.*

7.4 KAM conditions

By the KAM theorem applied to Hamiltonian systems near periodic orbits [5], an orbit is non-linearly stable if:

1. All non-trivial Floquet multipliers lie on the unit circle (linear stability).
2. All Floquet angles θ_j satisfy non-resonance conditions $\sum_j n_j\theta_j \neq 2\pi k$ for all integer vectors (n_j) with $\sum_j |n_j| \leq 4$ and integers k.
3. A non-degeneracy (twist) condition holds for the Birkhoff normal form to fourth order.

Conditions (1) and (2) are checkable from the Floquet spectrum; condition (3) requires computing the fourth-order Birkhoff normal form, which is numerically intensive but tractable for small N.

8 Implications for the Research Programme

The mathematical analysis above has several direct consequences for the classification programme:

1. **Non-planarity is a symmetry statement.** Whether a given point group yields 3D orbits is determined exactly by Propositions 4.1–4.4: the z-modes survive iff the group does not contain an element that eliminates them. The summary table in Section 4 directly guides which groups to search.
2. **Collision-free search is guaranteed for odd C_n and C_{nv}.** Proposition 5.2 means the softening regularisation used in computation can be removed for these families without risk of spurious near-collision minimisers.
3. **The knot type is constrained by the symmetry group.** The conjectured $T(2, n)$ torus knot structure for C_n choreographies, if confirmed numerically, would constitute a new theorem. Its proof would use the symmetry-constrained form of Γ and the classification of periodic braids under cyclic group actions.
4. **The CompSpec is revised.** The Reynolds projector construction in the CompSpec should now use the explicit mode selection rules derived here (Section 4) rather than the generic Reynolds averaging, which is equivalent but far more expensive to compute. The mode selection table allows P_G to be constructed in $O(K)$ rather than $O(|G| \cdot K)$ time.
5. **The RDD claim on icosahedral tractability is justified.** The "dimensional miracle" noted in Section 4.6 (the search space dimension at fixed K/N is

independent of $|G|$) formally justifies the optimistic treatment of icosahedral targets in the RDD.

References

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